

JOHN ABBOUD, C.E.T.

GIS Technologist in Training - Bachelor of Applied
Technology, Geographic Information Systems, SAIT

Calgary, AB · 403-477-3771 · john.abboud@proton.me ·
[linkedin.com/in/abboud-john](https://www.linkedin.com/in/abboud-john)



john-abboud.github.io

GIS Student currently completing SAIT's Bachelor of Applied Technology in Geographic Information Systems (BGIS) with applied experience in composite spatial index development, PostgreSQL/PostGIS database implementation, spatial statistics, and enterprise/web GIS development for a capstone client project. Skilled in ArcGIS Pro, ArcGIS Online, Python, SQL, and FME. Brings a C.E.T. designation and over 15 years of prior experience in electronics and instrumentation across aerospace, oilfield, and utility industries, contributing a strong foundation in precision troubleshooting, root-cause analysis, and documentation that carries directly into GIS data quality and field work.

GIS EXPERIENCE

GIS Analyst - Student (Semester 1 Final Project) | SAIT, Calgary (Jan 2026 - Apr 2026)

Responsibilities

- Co-developed "Where the Sun Shines and Income Aligns," a GIS-based rooftop solar suitability analysis for Calgary, integrating solar radiation, residential rooftop area, and median household income into a composite suitability index.
- Built a solar radiation raster surface from LiDAR-derived DEM data and applied zonal statistics, spatial joins, and Esri Enrich demographic data at the community level.
- Designed and implemented a parallel PostgreSQL/PostGIS spatial database with an Entity Relationship Diagram, foreign key constraints, and advanced spatial SQL queries.
- Conducted spatial statistical analysis, including Hot Spot Analysis (Getis-Ord G_i^*), Cluster and Outlier Analysis (Local Moran's I), and Global Moran's I , and produced bivariate choropleth mapping of solar potential against income.
- Automated the end-to-end workflow using ArcGIS ModelBuilder, from data cleaning and topology validation through composite index calculation.

Achievements

- Identified and ranked the top Calgary communities for rooftop solar development (Edgemont, Tuscany, Springbank Hill, Aspen Woods, among others), validated across two independent methods (composite index and bivariate analysis).
- Delivered a 68-page technical report including full data dictionary, QA/QC checklist, and PostGIS query appendix.

GIS Analyst - Student (Capstone Project) | CIRUS Lab / SAIT, Calgary (May 2026 - Aug 2026)

Responsibilities

- Co-developed "Highway to the Trigger Zone," an avalanche trigger-zone digital twin for Morrissey Ridge, BC (72.5 km²), for the CIRUS Lab at SAIT, applying aviation/RPAS domain expertise within a 3-person team.
- Built a six-factor weighted suitability model (slope, plan curvature, aspect, ruggedness, elevation, land cover) from 1m LiDAR DEM/DSM data using an AHP weighting scheme, automated through a 3-tool ArcPy workflow.
- Led delivery of the Experience Builder digital twin, integrating 2D/3D terrain review, weather context, field validation apps (Field Maps/Survey123/QuickCapture), and an ArcGIS Dashboard into a single decision-support application.
- Applied RPAS/aviation domain knowledge to a two-run drone route planning tool, deliberately keeping final target selection with the qualified human operator rather than automating it.

Achievements

- Model identified 268 candidate trigger zones (406.6 ha) narrowed from a 72.5 km² study area, validated by transferring the unchanged workflow to an independent test area and comparing results against professional ATES terrain classifications.
- Delivered a working recorded demonstration of the two-run route planner and a print-ready mission map export requested directly by the CIRUS Lab client.

ADDITIONAL GIS PROJECTS

- Automated Python/ArcPy geoprocessing scripts - cycling-network analysis by township (Toulouse, France) and ATS/DLS land-fabric import, clip, and GPS-point location workflow (Alberta).
- Supervised multispectral land cover classification (Sibbald Lake, AB) - 84.5% overall accuracy, supporting ecological sampling design.
- Thematic cartography - choropleth, proportional symbol, and data-classification method comparison maps (GEOS 457).

EXTRACURRICULAR LEADERSHIP

President | SAIT BGIS Club (Apr 2026 - Aug 2026)

- Coordinated promotion of GIS-related events and initiatives.
- Supported communication between students, instructors, and industry partners.
- Strengthened ability to explain GIS concepts and outcomes to diverse audiences.

CERTIFICATION & MEMBERSHIP

- Certified Engineering Technologist (C.E.T.) - ASET, since May 2011
- Student Member - Alberta Society of Surveying and Mapping Technologies (ASSMT), since Feb 2026
- Small Remotely Piloted Aircraft (VLOS) Basic Operations - Transport Canada (Jul 2026)
- Integrate Data with the FME Platform - Safe Software FME Academy (Aug 2026)
- H2S Alive (Energy Safety Canada, expires May 2027) and Standard First Aid/CPR/AED Level C (Canadian Red Cross, expires Jan 2027)

PROFESSIONAL EXPERIENCE (PRE-GIS CAREER)

ILI Field Technician | United Pacific Projects, Calgary, AB (Jun 2024 - Jan 2025)

- Performed infield and shop-based maintenance, repairs, and cleaning on pipeline in-line inspection tools.
- Travelled to remote sites to prepare inspection tools for surveys and advised on field conditions affecting data quality.

Electronics Repair Technician | Arcfield Canada, Calgary, AB (Jul 2019 - Apr 2024)

- Tested, repaired, and maintained CF-18 avionics systems to component level using automated test equipment and approved repair schemes.
- Calibrated test equipment to OEM specifications and maintained documentation under ISO 9001/AS9100 and Airworthiness standards; maintained full calibration traceability with zero documentation non-conformances.

Wireline Electronics Technologist III | Weatherford Canada Partnership, Grande Prairie / Red Deer, AB (Jun 2009 - Mar 2019)

- Diagnosed and repaired electronic and mechanical faults in wireline tools; performed preventive, corrective, and calibration maintenance.
- Participated in root cause analysis, trained new hires, and managed spare parts inventory and on-call support; recognized as a go-to technician for complex field diagnostics over a 9-year tenure.

EDUCATION

Bachelor of Applied Technology, Geographic Information Systems | SAIT, Calgary, AB (2025 - Present)

- Coursework in cartographic design, spatial analysis, geodatabase design, GPS field data capture, remote sensing/classification, and geoprocessing automation (Python/ArcPy, FME, ModelBuilder).

Electronics Engineering Technology Diploma | SAIT, Calgary, AB (2009)

- TAC-accredited program: circuit design and analysis, digital/analog electronics, microprocessor systems, wireless communications, and automated test and measurement; completed a capstone engineering design project.

TECHNICAL SKILLS

GIS Software

ArcGIS Pro, ArcGIS Enterprise, ArcGIS Online, QGIS, ArcGIS StoryMaps, Survey123, Experience Builder, Field Maps, QuickCapture, Dashboards

Spatial Analysis

Coordinate systems & projections, geodatabase design, spatial joins/queries, remote sensing & classification, cartographic design

Programming & Data

Python (ArcPy), SQL, PostgreSQL/PostGIS, FME, ModelBuilder, Excel

Professional & Language

Troubleshooting & root-cause analysis, technical documentation, multidisciplinary collaboration, English & Arabic (fluent), French (advanced)